

Calculus

1. Limits

1.1. Indeterminate Forms

- $\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty^0, \infty - \infty, 1^\infty, 0^0$

1.2. Properties of Limits

If $u = f(x)$ and $v = g(x)$,

- $\lim_{x \rightarrow a} [u + v] = \lim_{x \rightarrow a} u + \lim_{x \rightarrow a} v$
- $\lim_{x \rightarrow a} [u - v] = \lim_{x \rightarrow a} u - \lim_{x \rightarrow a} v$
- $\lim_{x \rightarrow a} [u \cdot v] = \lim_{x \rightarrow a} u \cdot \lim_{x \rightarrow a} v$
- $\lim_{x \rightarrow a} \left[\frac{u}{v} \right] = \frac{\lim_{x \rightarrow a} u}{\lim_{x \rightarrow a} v}$
- $\lim_{x \rightarrow a} [v]^n = \left[\lim_{x \rightarrow a} v \right]^n$

1.3. Evaluating Limits

- $\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n} = \left(\frac{m}{n} \right) \cdot a^{m-n}$
- $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$
- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

- $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$
- $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$
- L'Hôpital's Rule: If the limit is in $\frac{0}{0}$ or $\frac{\infty}{\infty}$ indeterminate form,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

2. Differentiation

2.1. First Principle

- $\frac{d}{dx}(f(x)) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

2.2. Common Properties

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$
- $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$

2.3. Product and Quotient Rule

If $u = f(x)$ and $v = g(x)$,

- $\frac{d}{dx}(u \cdot v) = u \cdot \frac{d}{dx}(v) + v \cdot \frac{d}{dx}(u)$
- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\left[v \cdot \frac{d}{dx}(u)\right] - \left[u \cdot \frac{d}{dx}(v)\right]}{v^2}$

2.4. Chain Rule

- $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

2.5. Power Rule

- $\frac{d}{dx}(x^n) = nx^{n-1}$

2.6. Common Derivatives

1. $\frac{d}{dx}(x) = 1$
2. $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$
3. $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$
4. $\frac{d}{dx}(a^x) = a^x \ln a$
5. $\frac{d}{dx}(e^x) = e^x$
6. $\frac{d}{dx}(e^{mx}) = me^{mx}$
7. $\frac{d}{dx}(\ln|x|) = \frac{1}{x}$
8. $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$

2.7. Derivatives of Trig Functions

1. $\frac{d}{dx}(\sin x) = \cos x$
2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\tan x) = \sec^2 x$
4. $\frac{d}{dx}(\cot x) = -\csc^2 x$
5. $\frac{d}{dx}(\sec x) = \sec x \tan x$
6. $\frac{d}{dx}(\csc x) = -\csc x \cot x$

2.8. Derivatives of Inverse Trig Functions

1. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
2. $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$
3. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
4. $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$
5. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
6. $\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$

2.9. Derivatives of Hyperbolic Functions

1. $\frac{d}{dx}(\sinh x) = \cosh x$
2. $\frac{d}{dx}(\cosh x) = \sinh x$
3. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
4. $\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$
5. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$
6. $\frac{d}{dx}(\operatorname{csch} x) = -\operatorname{csch} x \coth x$

2.10. Derivatives of Inverse Hyperbolic Functions

1. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2 + 1}}$
2. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$
3. $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1 - x^2}$
4. $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1 - x^2}$
5. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1 - x^2}}$
6. $\frac{d}{dx}(\operatorname{csch}^{-1} x) = \frac{-1}{|x|\sqrt{1 + x^2}}$

3. Integration

3.1. Common Properties

- $\int c dx = cx + C$
- $\int c \cdot f(x) dx = c \int f(x) dx$
- $\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$

3.2. Power Rule

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

3.3. Common Integrals

1. $\int \frac{1}{x} dx = \ln|x| + C$
2. $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$
3. $\int e^x dx = e^x + C$
4. $\int e^{mx} dx = \frac{1}{m} e^{mx} + C$
5. $\int a^x dx = \frac{a^x}{\ln a} + C$
6. $\int \ln x dx = x \ln x - x + C$

3.4. Integrals of Trig Functions

1. $\int \sin x dx = -\cos x + C$

2. $\int \cos x dx = \sin x + C$
3. $\int \tan x dx = \ln|\sec x| + C$
4. $\int \cot x dx = \ln|\sin x| + C$
5. $\int \sec x dx = \ln|\sec x + \tan x| + C$
6. $\int \csc x dx = \ln|\csc x - \cot x| + C = \ln\left|\tan \frac{x}{2}\right| + C$
7. $\int \sec^2 x dx = \tan x + C$
8. $\int \csc^2 x dx = -\cot x + C$
9. $\int \sec x \tan x dx = \sec x + C$
10. $\int \csc x \cot x dx = -\csc x + C$

3.5. Integrals of Inverse Trig Functions

1. $\int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1 - x^2} + C$
2. $\int \cos^{-1} x dx = x \cos^{-1} x - \sqrt{1 - x^2} + C$
3. $\int \tan^{-1} x dx = x \tan^{-1} x - \frac{1}{2} \ln|1 + x^2| + C$
4. $\int \cot^{-1} x dx = x \cot^{-1} x + \frac{1}{2} \ln|1 + x^2| + C$
5. $\int \sec^{-1} x dx = x \sec^{-1} x - \ln|x + \sqrt{x^2 - 1}| + C$
6. $\int \csc^{-1} x dx = x \csc^{-1} x + \ln|x + \sqrt{x^2 - 1}| + C$

3.6. Integrals of Hyperbolic Functions

1. $\int \sinh x dx = \cosh x + C$
2. $\int \cosh x dx = \sinh x + C$
3. $\int \tanh x dx = \ln|\cosh x| + C$
4. $\int \coth x dx = \ln|\sinh x| + C$
5. $\int \operatorname{sech} x dx = \tan^{-1}(\sinh x) + C$
6. $\int \operatorname{csch} x dx = \ln\left|\tanh \frac{x}{2}\right| + C$

3.7. Integrals of Inverse Hyperbolic Functions

1. $\int \sinh^{-1} x dx = x \sinh^{-1} x - \sqrt{x^2 + 1} + C$
2. $\int \cosh^{-1} x dx = x \cosh^{-1} x - \sqrt{x^2 - 1} + C$
3. $\int \tanh^{-1} x dx = x \tanh^{-1} x + \frac{1}{2} \ln|1 - x^2| + C$
4. $\int \coth^{-1} x dx = x \coth^{-1} x + \frac{1}{2} \ln|x^2 - 1| + C$
5. $\int \operatorname{sech}^{-1} x dx = x \operatorname{sech}^{-1} x + 2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} + C$
6. $\int \operatorname{csch}^{-1} x dx = x \operatorname{csch}^{-1} x + \sinh^{-1}|x| + C$

3.8. Advanced Integrals

1. $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

$$\begin{aligned}
2. \quad & \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C \\
3. \quad & \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C \\
4. \quad & \int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left| x + \sqrt{x^2 + a^2} \right| + C = \sinh^{-1} \frac{x}{a} + C \\
5. \quad & \int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left| x + \sqrt{x^2 - a^2} \right| + C = \cosh^{-1} \frac{x}{a} + C, (x > a) \\
6. \quad & \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C \\
7. \quad & \int \sqrt{x^2 + a^2} dx = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C \\
& \quad \quad \quad = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \sinh^{-1} \frac{x}{a} + C \\
8. \quad & \int \sqrt{x^2 - a^2} dx = \frac{x\sqrt{x^2 - a^2}}{2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + C \\
& \quad \quad \quad = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \cosh^{-1} \frac{x}{a} + C \\
9. \quad & \int \sqrt{a^2 - x^2} dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C
\end{aligned}$$

3.9. Definite Integrals

- $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$
- $\int_a^b f(x) dx = - \int_b^a f(x) dx$
- $\int_a^b f(x) dx = \int_a^b f(z) dz$
- $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$
- $\int_a^b f(x) dx = \int_{a-c}^{b-c} f(x+c) dx$

- $\int_a^b f(x)dx = \int_{a+c}^{b+c} f(x-c)dx$
- $\int_a^\infty f(x)dx = \lim_{b \rightarrow \infty} \int_a^b f(x)dx$

4. Solving Integrals

4.1. u -Substitution

- Formula:

$$\int f(g(x)) \cdot g'(x)dx = \int f(u) \cdot du \quad \text{where } u = g(x), du = g'(x)dx$$

- Definite Integrals:

$$\int_a^b f(g(x)) \cdot g'(x)dx = \int_{g(a)}^{g(b)} f(u) \cdot du$$

4.2. Integration by Parts

- Standard Form:

$$\int u dv = uv - \int v du$$

- Expanded Form:

$$\int u \cdot v dx = u \int v dx - \int \left[\frac{d}{dx}(u) \int v dx \right] dx$$

- LIATE Priority Rule for u :

- Logarithmic ($\ln x$)
- Inverse Trig ($\tan^{-1} x$)
- Algebraic ($x^2, 3x$)
- Trigonometric ($\sin x$)
- Exponential (e^x)

4.3. Trigonometric Substitution

- $\sqrt{a^2 - x^2} \Rightarrow x = a \sin \theta$
- $\sqrt{a^2 + x^2} \Rightarrow x = a \tan \theta$
- $\sqrt{x^2 - a^2} \Rightarrow x = a \sec \theta$

4.4. Weierstrass Substitution (Tangent Half-Angle)

Used for rational functions of $\sin x$ and $\cos x$. Let $t = \tan\left(\frac{x}{2}\right)$:

- $dx = \frac{2dt}{1+t^2}$
- $\sin x = \frac{2t}{1+t^2}$
- $\cos x = \frac{1-t^2}{1+t^2}$
- $\tan x = \frac{2t}{1-t^2}$

4.5. Shortcuts & Special Integrals

- $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$
- $\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)} + C$
- $\int e^{mx}[mf(x) + f'(x)]dx = e^{mx}f(x) + C$
- $\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2}(a \sin(bx) - b \cos(bx)) + C$
- $\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2}(a \cos(bx) + b \sin(bx)) + C$

4.6. Reduction Formulas

- $\int \sin^n x dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x dx$
- $\int \cos^n x dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx$
- $\int \sec^n x dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x dx$

5. Sequences & Series

5.1. Notations

- Sequence Notation:

$$\{a_i\}_{i=k}^n$$

- i : Index variable
- k : Lower limit (default: 1)
- n : Upper limit (default: ∞)
- $i, k, n \in \mathbb{Z}^+$

- Series Notation:

$$\sum_{i=k}^n a_i$$

- i : Index variable
- k : Lower limit (default: 1)
- n : Upper limit (default: ∞)
- $i, k, n \in \mathbb{Z}^+$

5.2. Common Series

- Geometric Series:

$$\sum_{i=1}^{\infty} ar^{i-1} = a + ar + ar^2 + \dots$$

$$|r| < 1$$

- Converges: $(-1 < r < 1)$ with sum $S = \frac{a}{1-r}$
- Diverges: $|r| \geq 1$

- P-Series:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \dots$$

- Converges: $p > 1$
- Diverges: $p \leq 1$
- $f(x) = \frac{1}{x^p}$ is positive, continuous, and decreasing on $[1, \infty)$

- Harmonic Series:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

- Special P-Series where $p = 1$ (Always Diverges)

- Alternating Series:

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n = a_1 - a_2 + a_3 - a_4 + \dots$$

- Converges if:
 - $a_{n+1} \leq a_n$ for all n (Decreasing)
 - $\lim_{n \rightarrow \infty} a_n = 0$

- Alternating Harmonic Series:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

- Special Alternating Series where $a_n = \frac{1}{n}$ (Always Converges)

- Telescoping Series:

$$\sum_{n=k}^{\infty} (a_n - a_{n+1}) = a_k - \lim_{n \rightarrow \infty} a_{n+1}$$

- Power Series:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + \dots$$

- Centered at origin ($c=0$): $\sum_{n=0}^{\infty} a_n x^n$
- Defines a function $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$ on its domain of convergence.
- Radius and Interval of Convergence:
 - Converges only at $x=c$ ($R=0$)
 - Converges for all x ($R=\infty$)
 - Converges for $|x-c| < R$ and diverges $|x-c| > R$

Note

Endpoints $x=c-R$ and $x=c+R$ must be tested individually using other convergence tests.

- Calculus of Power Series:
 1. Differentiation:

$$f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$$

$$f'(x) = \sum_{n=1}^{\infty} n a_n(x-c)^{n-1}$$

1. Integration:

$$f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n \quad \int f(x)dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-c)^{n+1}$$

1. Key Radius Rule: Differentiation and integration do not change the radius of convergence (R), but they can change the convergence at the endpoints ($x=c \pm R$).

5.3. Convergence Tests

- Divergence Test:

$$\lim_{n \rightarrow \infty} a_n \neq 0 \text{ or DNE} \Rightarrow \text{Diverges}$$

$$\lim_{n \rightarrow \infty} a_n = 0 \Rightarrow \text{Inconclusive}$$

- Direct Comparison Test: For positive terms ($a_n, b_n > 0$):
 - If $a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.
 - If $a_n \geq b_n$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.
- Limit Comparison Test: For positive terms ($a_n, b_n > 0$):

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$$

- If $0 < L < \infty$, both series share convergence status.
 - If $L = 0$ and $\sum b_n$ converges, $\sum a_n$ converges.
 - If $L = \infty$ and $\sum b_n$ diverges, $\sum a_n$ diverges.
- Absolute Convergence Test: If $\sum |a_n|$ converges, $\sum a_n$ converges (Absolute convergence implies convergence).
- Ratio Test:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \Rightarrow \text{Absolute Convergence}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1 \Rightarrow \text{Divergence}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \Rightarrow \text{Inconclusive}$$

- Root Test:

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1 \Rightarrow \text{Absolute Convergence}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1 \Rightarrow \text{Divergence}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1 \Rightarrow \text{Inconclusive}$$

- Integral Test: For $f(n) = a_n$ continuous, positive, and decreasing on $[k, \infty)$:

$$\sum_{n=k}^{\infty} a_n \text{ and } \int_k^{\infty} f(x) dx \text{ share convergence status.}$$

5.4. Absolute vs Conditional Convergence

- Absolute Convergence: $\sum a_n$ and $\sum |a_n|$ both converges.
- Conditional Convergence: $\sum a_n$ converges but $\sum |a_n|$ diverges (e.g. Alternating Harmonic Series).

5.5. Taylor & Maclaurin Series

- Taylor Series:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

- Maclaurin Series:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

5.6. Evaluated Taylor Series

$$1. e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$2. e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$3. a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \dots$$

$$4. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (-1 < x \leq 1)$$

$$5. \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \quad (-1 \leq x < 1)$$

$$6. \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$7. \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$8. \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad (-1 \leq x \leq 1)$$

$$9. (1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \dots \quad (-1 < x < 1)$$

5.7. Remainder & Error Estimation

- Definition of Remainder (R_n): If $S = \sum_{k=1}^{\infty} a_k$ converges with partial sum $S_n = \sum_{k=1}^n a_k$, the error (remainder) is:

$$R_n = S - S_n = \sum_{k=n+1}^{\infty} a_k \quad \text{where} \quad \lim_{n \rightarrow \infty} R_n = 0$$

- Alternating Series Estimation Theorem: For a convergent alternating series, the error is bounded by the first omitted term:

$$|R_n| = |S - S_n| \leq a_{n+1}$$

- Finding n for a desired accuracy ε : Set the error bound strictly less than ε and solve for n :

$$a_{n+1} < \varepsilon$$

- Integral Test Error Bound: If $f(x) = a_n$ is continuous, positive, and decreasing for $x \geq n$:

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$$

- Finding n for a desired accuracy ε : Use the upper bound to ensure the error stays below ε :

$$\int_n^{\infty} f(x) dx < \varepsilon$$

- Taylor Series Remainder (Lagrange Error Bound): If $f(x)$ is approximated by its n -th degree Taylor polynomial $T_{n(x)}$ centered at a :

$$R_{n(x)} = f(x) - T_{n(x)} = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

where c lies between a and x .

- Finding n for a desired accuracy ε : Find $M = |f^{(n+1)}(c)|$ on the interval between a and x , then solve:

$$\left| R_{n(x)} \right| \leq \frac{M}{(n+1)!} |x-a|^{n+1} < \varepsilon$$